

## Lab Exercise #6: Multiplying Square Matrices

## 1 Overview

In mathematics a *matrix* is a rectangular array of numbers. Matrices are named with capital letters and the size of a matrix is given by  $m \times n$  where  $m$  is the number of rows and  $n$  is the number of columns. When  $m = n$  the matrix is called a *square* matrix. In the examples below,  $A$  is a  $3 \times 4$  matrix and  $B$  is square.

$$A = \begin{bmatrix} 1 & -36 & -94 & 52 \\ -41 & 23 & 33 & 9 \\ 4 & -8 & 62 & 11 \end{bmatrix} \qquad B = \begin{bmatrix} 0 & -8 & 14 \\ 20 & 1 & 22 \\ -7 & 12 & 3 \end{bmatrix}$$

In order to multiply two matrices,  $A_{m \times n} \times B_{t \times s}$ , the values of  $n$  and  $t$  must be the same. The result is a matrix with size  $m \times s$ . To multiply the matrices, multiply the elements of a row  $i$  in  $A$  by the corresponding elements of column  $j$  in  $B$  and add them together. This result goes in  $c_{ij}$  of the resulting matrix. More information may be found at

Consider the example given below.

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & 6 \\ 7 & 8 \end{bmatrix} = \begin{bmatrix} 1(5) + 2(7) & 1(6) + 2(8) \\ 3(5) + 4(7) & 3(6) + 4(8) \\ 0(5) + 1(7) & 0(6) + 1(8) \end{bmatrix} \\ = \begin{bmatrix} 19 & 22 \\ 43 & 50 \\ 7 & 8 \end{bmatrix}$$

In this lab, write a C program that multiplies two square matrices ( $5 \times 5$  or smaller) and outputs the resulting matrix.

Your program should

- Request the user to input the number of rows in the matrices and prompt the user to input the matrices. [**Note:** It is not required to input the first two matrices on separate lines. I did that for readability.]
- Using a nested loop, calculate the product of the matrices.
- Using a nested loop, output the resulting matrix.

## 2 Code Documentation

The very first lines in *all* of your programming project files this semester should be comments in the class header file (copy from header.c). Of course, the items in the square brackets [] in the comments should be filled in with the proper information.

In addition, the first output for all of your projects this semester, both for the lab exercises and the programming assignments, should be the name of the project and your name, e.g:

```
printf("Lab #6 - Joe Student\n");
```

Obviously, for the lab exercises, the output should include the names of both students in the lab group, e.g:

```
printf("Lab #6 - Joe Student and Jane Doe\n");
```

## 3 Sample Run

Your output should match the sample run below.

```
Lab #7: Mark Schwitzerlett and Jacobi

Enter the number of rows, N, of a square matrix, (N <= 5): 4

Enter the 16 elements of Matrix-1:
11 8 -3 4
6 1 0 7
4 -23 12 4
-5 3 20 -18

Enter the 16 elements of Matrix-2:
-1 2 3 4
19 8 -7 5
4 3 -2 0
12 -10 40 3

The product of the two matrices is:
177      37      143      96
97      -50      291      50
-345     -180      309     -87
-74      254     -796     -59

...Program finished with exit code 0
Press ENTER to exit console.[]
```

## 4 Deliverables

You should turn in a stand-alone, complete application program (your C source code, a single “.c” file) containing a `main()` function. All labs and projects this semester will be submitted in Gradescope (via a Canvas assignment).

Your file should be named `Lab#6_FirstName_LastName.c`

**Due date:** The lab is due at the end of your lab period, either Monday, March 23rd or Tuesday, March 24th.

**Be sure to save a copy of your .c file.**